3.3.3 Pregenerated SVG

SVG has been around for a while, and there are, not surprisingly, robust tools for drawing SVG, like Adobe Illustrator and the open source tool Inkscape. You'll likely want pregenerated SVG for icons, interface elements, and other components of your work. If you're interested in icons, The Noun Project (http://thenounproject.com/) has an extensive repository of SVG icons, including the football in figure 3.20.

When you download an icon from The Noun Project, you get it in two forms: SVG and PNG. You've already learned how to reference images, and you can do the same with SVG by pointing the `xlink:href` attribute of an `<image>` element at an SVG file. But loading SVG directly into the DOM gives you the capacity to manipulate it like any SVG elements that you create in the browser with D3.

Let's say we decide to replace our boring circles with balls, and we don't want them to be static images because we want to be able to modify their color and shape like other SVG. In that case, we'll need to find a suitable ball icon and download it. In the case of downloads from The Noun Project, this means we'll need to go through the hassle of creating an account, and we'll need to properly attribute the creator of the icon or pay a fee to use the icon without attribution. Regardless of where we get our icon, we might need to modify it before using it in our data visualization. In the case of the football icon in this example, we need to make it smaller and center the icon on the 0,0 point of the canvas. This kind of preparation is going to be different for every icon, depending on how it was originally drawn and saved.



Figure 3.20 An icon for a football created by James Zamyslianskyj and available at http://thenounproject.com/term/football/1907/ from The Noun Project